Kevin **Tongue**

Double Degree – Mechanical Engineering & Master in Numerical Solid Mechanics

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Education

Centrale Lyon – ENISE & École Nationale Supérieure Polytechnique de Yaoundé (ENSPY) — *Saint-Étienne, France / Yaoundé, Cameroon*

Double Degree – Engineering in Mechanical Engineering & Master 2 in Numerical Solid Mechanics — *2020 – 2026 (expected)*

- **Specialisation:** Mechanical design, advanced numerical simulation, and computational mechanics.
- **Key courses (Master):** Finite Element Method, Continuum Mechanics, Fracture Mechanics, Structural Dynamics, Nonlinear Mechanics, Uncertainty Quantification, Design of Experiments, Residual Stress Analysis, Fluid Dynamics, Heat Transfer, Materials Science.
- **Key courses (Engineering):** Solid Mechanics, Thermodynamics, Machine Design, Manufacturing Processes, System Dynamics, Signal Processing, Applied Mathematics.

Research & Professional Experience

IMT Mines Albi & IMT Mines Saint-Étienne — *Albi / Saint-Étienne, France*

Master Internship – Deterministic–Stochastic Coupling for Granular Segregation (DEM + Markov Chains) — *Mar. 2026 – Sep. 2026*

- Coupled Discrete Element Method (DEM) simulations with an inhomogeneous Markov chain model to statistically characterise granular segregation dynamics in mixing devices.
- Extracted particle-scale transition probabilities from DEM outputs; developed stochastic models linking microscopic particle dynamics to macroscopic distributions.
- Designed and executed a Design of Experiments (DoE) plan and performed Monte Carlo sensitivity analysis for uncertainty quantification (UQ).
- Implemented end-to-end numerical pipelines in Python (NumPy, SciPy, PyTorch, Matplotlib) including data fusion, signal processing, and model validation.

Centrale Lyon – ENISE — *Saint-Étienne, France*

Research Project – Nonlinear FEM Solver for Fatigue Failure Prediction (Dang Van Criterion) — *Oct. 2025 – Mar. 2026*

- Implemented a 3D nonlinear finite element solver in Python/C++ covering static, modal, and transient dynamic analyses.
- Applied and validated the Dang Van multiaxial fatigue criterion, including critical plane determination (Agard et al. optimisation methodology).
- Validated results against analytical benchmarks and Abaqus reference solutions; produced structured technical documentation.
- Deployed as a web application: `huggingface.co/spaces/ktongue/simulations_apps`

Centrale Lyon – ENISE (collaboration with LTDS Laboratory) — *Saint-Étienne, France*

Research Project – Physics-Informed Neural Networks (PINN) for Material Identification — *2025*

- Built a Physics-Informed Neural Network (PINN) in PyTorch coupling deep learning with physical constraints (elasticity, heat equation, plasticity criteria).
- Performed inverse identification of constitutive law parameters from compression test data provided by the LTDS laboratory; validated the identified law via FEM simulations.
- Implemented data-driven surrogate modelling pipeline: preprocessing, training loop, hyperparameter tuning, and result visualisation.
- Deployed at: `huggingface.co/spaces/ktongue/material_identification`

Centrale Lyon – ENISE — *Saint-Étienne, France*

Laboratory Coursework – Surface Integrity and Residual Stress Characterisation in Machining — *2023 – 2024*

- Measured residual stresses induced by machining operations using laser-based spectroscopy techniques, and correlated results with cutting parameters and tool wear state.
- Conducted experimental micro-hardness mapping (Vickers micro-indentation) on machined 42CrMo4 steel specimens, comparing surface and core hardness across new vs. worn cutting inserts and parallel vs. perpendicular cutting directions.
- Characterised microstructure and tool wear mechanisms using Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDXS) to identify coating degradation and material composition changes at cutting edges.
- Analysed the influence of machining operations on surface integrity (hardness gradients, microstructural alteration, residual stress state) to define recommendations ensuring part integrity.

École des Mines de Saint-Étienne — *Saint-Étienne, France*

Academic Project – Microstructure Partitioning Model (Image Segmentation) — *Feb. – Jun. 2025*

- Processed and segmented microstructural images using Python (NumPy, OpenCV, SciPy); performed geometric feature extraction and statistical characterisation of material phases.
- Developed modular segmentation and classification algorithms applicable to materials with heterogeneous microstructures.

Centrale Lyon – ENISE — *Saint-Étienne, France*

Academic Project – Design of a Food Industry Lifting Trolley — *Mar. – Jun. 2025*

- Pre-dimensioning study, strength of materials calculations, and 3D CAD modelling (SolidWorks); ran static finite element simulation (SolidWorks Simulation) to validate structural integrity.
- Produced full technical documentation including functional analysis, tolerancing, and manufacturing drawings.

École des Mines de Saint-Étienne — *Saint-Étienne, France*

Academic Project – Web Application for Library Management — *Feb. – Jun. 2025*

- Designed and developed a full-stack library management web application using Symfony (PHP); implemented MVC architecture, database design, and user authentication.

Centrale Lyon – ENISE — *Saint-Étienne, France*

Academic Project – Design of a Gardening Assistance Cart — *Sep. – Dec. 2024*

- Mechanical design and CAD modelling with SolidWorks and CATIA; functional analysis, ergonomic study, component selection, and technical documentation.

SOPECAM (Société de Presse et d'Éditions du Cameroun) — *Yaoundé, Cameroon*

Engineering Internship – Industrial Diagnostics & Preventive Maintenance — *May 2024 – Sep. 2024*

- Investigated root causes of pump failures on KBA newspaper printing machines; proposed corrective and preventive maintenance strategies.
- Designed a software package for managing the preventive maintenance schedule of printing press pumps.
- Instrumented machines with vibration and temperature sensors; performed signal processing and failure mode analysis (FMEA approach).

FabLab – École Nationale Supérieure Polytechnique de Yaoundé (ENSPY) — *Yaoundé, Cameroon*

Technician Internship – Design & Manufacturing of a 3-Axis CNC Machine — *May – Sep. 2023*

- Full CAD design (SolidWorks: assemblies, sheet metal, machining preparation) and manufacturing of a 3-axis CNC machine for prototyping and educational use.
- Structural sizing, component selection (linear guides, stepper motors, belt transmission), and assembly supervision.

Technical Skills

Mechanics & Materials	Continuum mechanics, fracture mechanics, fatigue (Dang Van, Goodman), thermo-elasticity, cohesive zo basics), residual stress analysis (laser spectroscopy), constitutive law identification, multi-scale modellin
Materials Characterisation	Vickers micro-indentation hardness testing, Scanning Electron Microscopy (SEM), Energy-Dispersive X-ray microstructure analysis, surface integrity assessment in machining.
Finite Element Methods	Abaqus (static, dynamic, thermal), SolidWorks Simulation, ANSYS Workbench (basics), Cast3M (basics), cu solver (Python/C++): static, modal, and transient analyses.
Discrete & Stochastic Methods	Discrete Element Method (DEM/LIGGGHTS), Markov chain modelling, Monte Carlo simulation, Design of E uncertainty quantification, surrogate modelling.
AI & Machine Learning	Physics-Informed Neural Networks (PINNs, PyTorch), inverse parameter identification, data-driven micros deep learning for scientific computing.
Scientific Programming	Python (NumPy, SciPy, PyTorch, Matplotlib, OpenCV, Pandas), C/C++, MATLAB (Simulink, Computer Vision toolboxes), Fortran (basics), Java, LaTeX, Git/GitHub.
CAD & Design	SolidWorks (assemblies, sheet metal, simulation), CATIA V5 (part design, surfacing, PLM), 3DExperience p
Signal Processing & Instrumentation	Sensors (accelerometers, thermocouples, IMU, encoders), data acquisition, vibration analysis, GR&R, Cp/ analysis.
Web & Software Development	Python-based web applications (Gradio, Hugging Face Spaces), Symfony/PHP (MVC web apps), basic HT
Languages	French (native), English (B2 – professional technical proficiency, TOEIC), German (beginner).

Certifications & Activities

Centrale Lyon – ENISE — *Saint-Étienne, France*

Core Member – Robotics Club — *Nov. 2024 – Present*

- Integration of sensors (IMU, encoders, temperature sensors), real-time data processing, firmware development, and collaborative project management.

2024–2025	**MATLAB Certified**, MATLAB Onramp, Simulink Onramp, Computer Vision Onramp, Power System Onramp	*MathWorks*
2024	**Google Advanced Data Analytics**, Professional Certificate – data analysis, Python, statistics, machine learning	*Google / Coursera*
2024	**IBM Data Analyst**, Professional Certificate – data analysis, visualisation, SQL, Python	*IBM / Coursera*

Open-Source & Deployed Projects

2026 **FEM Fatigue Solver**, Nonlinear 3D FEM solver with Dang Van fatigue criterion – Python/C++ backend, interactive web interface *huggingface.co/spaces/kt*

2025 **PINN Material Identification**, Physics-Informed Neural Network for inverse identification of constitutive law parameters from compression tests *huggingface.co/spaces/kt*

Additional Information

- **Availability:** September/October 2026 – full geographic mobility (France and internationally).
- **Research interests:** Computational mechanics, granular media, fracture and fatigue, inverse problems, scientific machine learning, non-destructive evaluation (NDE), multi-physics simulation.
- **Soft skills:** Scientific rigour, analytical mindset, research autonomy, fast learner, team player, problem-solving orientation.